

## 1 0. Prof's Blueprint (High Priority)

- **Errors:** abs/rel error, MAE, RMSE on discrete points.
- **Polynomial forms:** standard, nested (Horner), or centered; FLOP counts.
- **Horner:** evaluate  $p(c)$ ; synthetic division by  $ax - b$ ; derivative via repeated Horner.
- **Taylor:** construct poly at given center; truncation error integral & Lagrange form; bound on interval (Sandwich Thm); find  $n$  for tolerance; Maclaurin/Taylor series convergence.
- **Interpolation:** setup linear system + matrix form  $Xa = y$ ; Vandermonde det (why nonzero); over/underdetermined & free parameters; Lagrange basis  $\ell_i(x)$  & poly  $p(x)$ ; extrapolation vs interpolation; divided differences (1st/2nd order only, no full Newton); Chebyshev nodes, Runge phenomenon; Bézier curves (basic).
- MCQ + written section, 30 pts total. Bring 1 A4 cheat sheet (this one!).

## 2 1. Purpose of NA & Solution Types (Th.1–2)

**Why numerical methods?** Exact answers often *unavailable* or *impractical*: (a) no closed-form expression exists (e.g.  $\int e^{-x^2} dx$ ,  $\sin x = x$ ); (b) formula exists but is too slow/unstable to evaluate; (c) data only known at discrete measured points. **Definitions:** **exact solution** = satisfies equation exactly (symbolic); **closed-form/analytical** = explicit expression in known functions/constants; **approximate (numerical)** = value from an algorithm with controlled error, generally not equal to exact. **Locality:** every approximation is built to be good *near* a reference point/interval — Taylor about  $x_0$ , interpolation on nodes  $[x_0, x_n]$ . Quality degrades away from that neighborhood  $\Rightarrow$  **extrapolation is unreliable**.

## 3 2. Error Formulas

$$E_{abs} = |f(x) - \tilde{f}(x)|, \quad E_{rel} = \frac{|f(x) - \tilde{f}(x)|}{|f(x)|}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |f(x_i) - \tilde{f}(x_i)|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - \tilde{f}(x_i))^2}$$

**Trap:** RMSE penalizes large errors more (squares); MAE treats all equally. Round to 5 dp unless told otherwise.

## 4 3. Polynomial Forms

**Standard:**  $p_n(x) = a_n x^n + \dots + a_1 x + a_0$  — needs  $\sim 2n - 1$  mults,  $n$  adds (naive) or  $n(n+1)/2$  mults w/o precomputed powers.

**Nested (Horner):**  $p_n(x) = (\dots((a_n x + a_{n-1})x + a_{n-2})x + \dots)x + a_0$  —  $n$  mults,  $n$  adds. **Always fewer ops.**

**Centered at  $x_0$ :**  $p_n(x) = \sum b_k (x - x_0)^k$ . Get  $b_k$  via repeated synthetic division by  $(x - x_0)$ , or Taylor:  $b_k = p^{(k)}(x_0)/k!$ .

**Horner's Method (synthetic division)**

Divide  $p(x)$  by  $(x - c)$ : bring down  $a_n$ , each next coeff =  $a_k + c \cdot$  (prev result). Last value = remainder =  $p(c)$ .

|     |           |            |         |        |
|-----|-----------|------------|---------|--------|
|     | $a_n$     | $a_{n-1}$  | $\dots$ | $a_0$  |
| $c$ |           | $cb_{n-1}$ | $\dots$ | $cb_0$ |
|     | $b_{n-1}$ | $b_{n-2}$  | $\dots$ | $r$    |

- $p(x) = (x - c)q(x) + r$ ,  $r = p(c)$ ; bottom row (minus last) = coeffs of  $q(x)$ .
- Divide by  $(ax - b)$ : divide by  $(x - b/a)$  first (root  $c = b/a$ ), get  $q^*(x)$ ; true quotient  $q(x) = q^*(x)/a$ , same remainder  $r$ .
- **Derivative link:** run the tableau on  $p(x)$  to get  $q(x)$  &  $r = p(c)$ ; run *another* tableau on  $q(x)$  at same  $c \Rightarrow$  new remainder =  $p'(c)$ . (Repeat on next quotient for  $p''(c)/2!$ , etc.)
- **Trap:** two ways to get  $p'(1)$  — direct Horner on  $p'(x)$ , or double-tableau on  $p(x)$  — must match.

## 5 4. Taylor & Maclaurin

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x)$$

Maclaurin:  $x_0 = 0$ .

## Truncation Error $R_n(x)$

**Integral form:**

$$R_n(x) = \int_{x_0}^x \frac{(x-t)^n}{n!} f^{(n+1)}(t) dt$$

**Lagrange form:**

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

for some  $\xi$  between  $x_0$  and  $x$ . **Relationship:** Lagrange form = Mean Value Thm applied to the integral form (weighted avg of  $f^{(n+1)}$  over interval).

**Bounding  $R_n(x)$  on interval  $[a, b]$  (Sandwich Thm)**

1. Find  $M = \max_{\xi \in [a, b]} |f^{(n+1)}(\xi)|$  (often = bound on derivative amplitude, e.g. for  $\sin, \cos$  terms  $M \leq 3^{n+1}$  etc.)
2. Find max of  $|x - x_0|^{n+1}$  on  $[a, b]$  (largest distance from center).
3.  $|R_n(x)| \leq \frac{M \cdot d^{n+1}}{(n+1)!}$  where  $d = \max |x - x_0|$ .

**Finding  $n$  for tolerance  $|R_n| < \varepsilon$**

Set bound expression  $< \varepsilon$ , solve for smallest integer  $n$  (try increasing  $n$  until factorial growth beats  $d^{n+1}$  growth). Use  $n!$  growth dominates exponential — trial/error with a table is acceptable.

**"Largest  $x$ " type (Quiz 3 Q5 style)**

Fix  $n$ ; treat bound  $\frac{M d^{n+1}}{(n+1)!} < \varepsilon$  as inequality in  $x$  (since  $d = x - x_0$  here); solve for  $x$ .

**Series Convergence**

Taylor series =  $f(x) \iff \lim_{n \rightarrow \infty} R_n(x) = 0$ . Writing the infinite sum from the pattern does *not* by itself prove convergence — must separately verify  $R_n \rightarrow 0$  (true for  $\sin, \cos, e^x$  on all  $\mathbb{R}$  since  $M$  bounded,  $d^{n+1}/(n+1)! \rightarrow 0$ ).

**Maclaurin of  $\sin(3x)$**

$$\begin{aligned} \sin(3x) &= \sum_{k=0}^{\infty} \frac{(-1)^k (3x)^{2k+1}}{(2k+1)!} \\ &= 3x - \frac{27x^3}{6} + \frac{243x^5}{120} - \dots \end{aligned}$$

**Trap:** degree-8 Maclaurin poly of an odd function still stops at  $x^7$  term (even-degree coeffs are 0) — write poly *through* degree 8, don't force an  $x^8$  term.

## 6 5. Chain Rule / Derivative Practice

$[f(g(x))] = f'(g(x))g'(x)$ . Common combos:  $2 \sin x + e^x$ ;  $1 + \cos^2 x = 1 + (\cos x)^2 \Rightarrow$  derivative =  $-2 \cos x \sin x = -\sin(2x)$ . **Worked (Quiz 2/HW):**  $f = 2 \sin x + e^x$ ;  $f' = 2 \cos x + e^x$ , at  $\pi/2$ :  $2 \cos(\pi/2) + e^{\pi/2} = e^{\pi/2} \approx 4.810$ .  $f = \sin^2(2x)$ :  $f' = 2 \sin(2x) \cos(2x) \cdot 2 = 2 \sin(4x)$  (chain  $\times 2$  from  $2x$ ).  $f = \cos x - \sin x$ :  $f' = -\sin x - \cos x$ . **Trap:** evaluate derivatives at the *center*  $x_0$  before plugging into Taylor coeff  $f^{(k)}(x_0)/k!$  — don't leave symbolic. Useful values:  $\sin(\pi/2) = 1, \cos(\pi/2) = 0$ ;  $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$ ;  $\sin(-\pi/4) = -1/\sqrt{2}$ .

## 7 6. Interpolation Basics

**Interpolation:** passes exactly through given data points (local fit). **Approximation:** fits overall trend, need not pass through points (global fit, e.g. least squares). **Extrapolation:** evaluate poly *outside* node range — much less reliable than interpolation (points 4-5 pattern in quizzes: always flag extrapolation as less trustworthy).

## Degree-1 (Linear) Interpolation

Given  $(x_0, y_0), (x_1, y_1)$ :  $p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$ .

**Main Types of Interpolation (Th.8)**

- **Linear (piecewise):** connect consecutive points by line segments. Simple, cheap,  $C^0$  only (kinks at nodes). Good for monotone data; crude derivative info.
- **Global poly:** Lagrange/Newton/Vandermonde — one degree- $n$  poly through  $n+1$  pts;  $C^\infty$ ; **unique** for distinct nodes. High degree  $\Rightarrow$  ill-conditioned, Runge.
- **Cubic splines:** piecewise cubic with matched  $f, f', f''$  at nodes;  $C^2$  smooth, local updates, avoids Runge — best for many nodes.
- **Hermite:** matches  $f$  and  $f'$  at nodes ( $2n+2$  conditions  $\Rightarrow$  degree  $2n+1$ ).

**Key tradeoff:** more smoothness  $\Rightarrow$  more conditions  $\Rightarrow$  higher degree (less stable); local (spline) methods avoid Runge, global ones risk it.

## Interpolation Error Bound

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

$$|f(x) - p_n(x)| \leq \frac{M}{(n+1)!} \max_x \left| \prod_{i=0}^n (x - x_i) \right|$$

where  $\xi \in [x_{\min}, x_{\max}]$ ,  $M = \max |f^{(n+1)}|$ . **Uses:** bound error at a point; choose nodes to shrink product term — why Chebyshev (minimize  $\max |\prod|$ ) beat uniform; error = 0 at every node. Analogous to Taylor remainder with  $(x - x_0)^{n+1}$  replaced by  $\prod (x - x_i)$ .

## Vandermonde System

For  $n+1$  points, poly degree  $n$ :  $a_0 + a_1 x_i + \dots + a_n x_i^n = y_i$  for each  $i$ . Matrix form  $Xa = y$  where

$$X = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{pmatrix}$$

$$\det(X) = \prod_{0 \leq i < j \leq n} (x_j - x_i)$$

**Why nonzero:** interpolation nodes  $x_i$  are *distinct*  $\Rightarrow$  every factor  $(x_j - x_i) \neq 0 \Rightarrow \det \neq 0 \Rightarrow$  unique solution guaranteed.

**Free Parameters (over/underdetermined)**

# points =  $m+1$ , poly degree =  $n$ . If  $n > m$ : **underdetermined** — infinitely many interpolating polys; free parameters =  $n - m$  (extra coefficients not pinned down by the  $m+1$  equations). If  $n < m$ : **overdetermined** — generally no exact solution (unless data is consistent)  $\rightarrow$  use approximation (least squares) instead. **Trap:** "degree at most  $n$ " with only  $m+1 < n+1$  points  $\Rightarrow$  family has  $n-m$  free parameters — identify them explicitly as the highest-order coefficients.

## Lagrange Interpolation

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \quad p(x) = \sum_{i=0}^n y_i \ell_i(x)$$

Properties:  $\ell_i(x_j) = \delta_{ij}$  (1 if  $i = j$ , else 0); basis *depends only on nodes  $x_i$* , not on  $y_i$  or  $f$  — same  $\ell_i$  reused for any function on same nodes. **Why a basis (Th.12):**  $\ell_0, \dots, \ell_n$  are  $n+1$  polys of degree  $\leq n$  and are *linearly independent* (a combo  $\sum c_i \ell_i \equiv 0$  evaluated at node  $x_j$  gives  $c_j = 0$ ). Since  $\dim P_n = n+1$ , they span all polys of degree  $\leq n \Rightarrow$  they form a basis; any data  $(x_i, y_i)$  yields unique coefficients  $c_i = y_i$ . **Steps:** (1) list nodes; (2) build each  $\ell_i(x)$  as product of  $(x - x_j)/(x_i - x_j)$  over  $j \neq i$ ; (3) sum  $y_i \ell_i(x)$ ; (4) simplify/evaluate at target point; (5) compute abs/rel error vs known true value if given; (6) extrapolate = plug  $x$  outside node range into same  $p(x)$ .

## 8 7. Divided Differences & Newton (Th.13)

0th order:  $f[x_i] = f(x_i)$  (the node value itself).

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

General (order  $k$ ), denom = last — first node:

$$f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

## Newton Interpolation Formula (Th.13)

$$p_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

**Advantages vs Lagrange:** adding a node adds only *one* term (no basis recompute); nested (Horner-like) eval cheaper; incremental DD table. Lagrange pros: no table, basis reusable for *any*  $f$  on same nodes. (Per prof: quizzes need only 1st/2nd order DD; know formula & why Newton wins for incremental data.)

9 8. Grids, Runge, Chebyshev, Bézier

- **Uniform grid:** equally spaced nodes,  $x_i = a + ih$ ,  $h = (b - a)/n$ . **Non-uniform:** unequal spacing (e.g. Chebyshev).
- **Runge's phenomenon:** high-degree interpolation on *uniform* nodes over  $[-1, 1]$ -type intervals can oscillate wildly near endpoints as  $n$  increases. Fix: Chebyshev nodes or splines.
- **Chebyshev nodes** (roots of  $T_{n+1}$ ), on  $[-1, 1]$ :

$$x_k = \cos\left(\frac{2k+1}{2(n+1)}\pi\right), \quad k = 0, \dots, n$$

On  $[a, b]$ :  $x_k = \frac{a+b}{2} + \frac{b-a}{2}\cos\left(\frac{2k+1}{2(n+1)}\pi\right)$ . Cluster near endpoints; minimize max of  $|\prod_{i=0}^n (x - x_i)|$  (interpolation error term)  $\Rightarrow$  minimize Runge oscillation.

- **Bézier curves:** defined by control points; curve passes through *endpoint* control points only (not interior ones); used in computer graphics for smooth curve design; based on Bernstein polynomials as blending functions.

10 9. Quick Reference Values

$e \approx 2.71828$ ,  $\pi \approx 3.14159$ ,  $\sqrt{2} \approx 1.41421$ ,  $1/\sqrt{2} \approx 0.70711$ .  $n!$ :  $5! = 120$ ,  $6! = 720$ ,  $7! = 5040$ ,  $8! = 40320$ ,  $9! = 362880$ ,  $10! = 3628800$ .

11 10. Worked Mini-Examples

**Horner:**  $p(x) = 2x^3 - 3x^2 + 0x + 5$  at  $c = 2$   
Coeffs: 2, -3, 0, 5.

|   |   |    |   |   |
|---|---|----|---|---|
|   | 2 | -3 | 0 | 5 |
| 2 |   | 4  | 2 | 4 |
|   | 2 | 1  | 2 | 9 |

So  $p(2) = 9$ ; quotient  $q(x) = 2x^2 + x + 2$ ;  $p(x) = (x - 2)(2x^2 + x + 2) + 9$ . **For**  $p'(2)$ : run another tableau on  $q$ 's coeffs 2, 1, 2 at  $c = 2$ :

|   |   |   |    |
|---|---|---|----|
|   | 2 | 1 | 2  |
| 2 |   | 4 | 10 |
|   | 2 | 5 | 12 |

So  $p'(2) = 12$ .

**Lagrange: nodes** (0, 1), (1, 3), (2, 7), **find**  $p(1.5)$

$$\begin{aligned} \ell_0(x) &= \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2} \\ \ell_1(x) &= \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{x(x-2)}{-1} \\ \ell_2(x) &= \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2} \end{aligned}$$

$p(x) = 1 \cdot \ell_0(x) + 3 \cdot \ell_1(x) + 7 \cdot \ell_2(x)$ . At  $x = 1.5$ :  $\ell_0 = \frac{(0.5)(-0.5)}{2} = -0.125$ ,  $\ell_1 = \frac{(1.5)(-0.5)}{-1} = 0.75$ ,  $\ell_2 = \frac{(1.5)(0.5)}{2} = 0.375$ .  $p(1.5) = 1(-0.125) + 3(0.75) + 7(0.375) = -0.125 + 2.25 + 2.625 = 4.75$ .

**Lagrange + errors: arctan (Quiz 5)**

$f(x) = \arctan x$ , nodes 0, 1, 3 ( $y = 0, 0.785, 1.249$ ).

$$\begin{aligned} \ell_0(x) &= \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{(x-1)(x-3)}{3} \\ \ell_1(x) &= \frac{x(x-3)}{(1-0)(1-3)} = \frac{x(x-3)}{-2} \\ \ell_2(x) &= \frac{x(x-1)}{(3-0)(3-1)} = \frac{x(x-1)}{6} \end{aligned}$$

$p(x) = 0\ell_0 + 0.785\ell_1 + 1.249\ell_2$ . At  $x = 2$ :  $\ell_1 = \frac{2(-1)}{-2} = 1$ ,  $\ell_2 = \frac{2(1)}{6} = \frac{1}{3} \Rightarrow p(2) = 0.785 + \frac{1.249}{3} = 1.20133$ .

**Errors** vs  $\arctan(2) \approx 1.10715$ :  $E_{abs} = |1.10715 - 1.20133| = 0.09418$ ,  $E_{rel} = \frac{0.09418}{1.10715} = 0.08507$  (8.5%).

**Extrapolate**  $x = -0.5$  (outside  $[0, 3]$ ):  $p(-0.5) = 0.785\ell_1(-0.5) + 1.249\ell_2(-0.5) = -0.53075$  (true  $\arctan(-0.5) = -0.46365$  — note large error: extrapolation unreliable).

**Divided Differences: nodes** (0, 1), (1, 3), (2, 7)

$f[x_0] = 1$ ,  $f[x_1] = 3$ ,  $f[x_2] = 7$ ;  $f[x_0, x_1] = \frac{3-1}{1-0} = 2$ ,  $f[x_1, x_2] = \frac{7-3}{2-1} = 4$ ;  $f[x_0, x_1, x_2] = \frac{4-2}{2-0} = 1$ . **Newton form:**  $p_2(x) = 1 + 2x + 1 \cdot x(x - 1) = 1 + 2x + x^2 - x^2 + x + 1$ . **Check:** same poly as Lagrange (uniqueness). At  $x = 1.5$ :  $2.25 + 1.5 + 1 = 4.75$  ✓ matches.

**Vandermonde det: nodes** 0, 1, 2

$$\det(X) = \prod_{0 \leq i < j \leq 2} (x_j - x_i) = (1-0)(2-0)(2-1) = 1 \cdot 2 \cdot 1 = 2 \neq 0$$

**Degree-1 Interp:**  $\{(2, 5), (5, -1)\}$  (Quiz 4 Q1)

$p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) = 5 + \frac{-1 - 5}{5 - 2}(x - 2) = 5 - 2(x - 2) = -2x + 9$ . Check:  $p_1(2) = 5$ ,  $p_1(5) = -1$ . ✓ (Also: slope  $m = -2$  through both pts.)

**Degree-4 System & det: 5 pts (Quiz 4 Q2)**

Points  $\{(0, 3), (1, 2), (2, 5), (4, 9), (7, 8)\}$ , degree 4:  $p_4(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ . **System**  $Xa = y$ , rows = points, cols = powers 0..4:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 4 & 16 & 64 & 256 \\ 1 & 7 & 49 & 343 & 2401 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 5 \\ 9 \\ 8 \end{pmatrix}$$

**det:**  $\det X = \prod_{i < j} (x_j - x_i)$  over the 5 nodes (10 ordered pairs  $i < j$ ):

$\det X = (1)(2)(1)(4)(3)(2)(7)(6)(5)(3) = 30240 \neq 0$

**Solving:** first eq gives  $a_0 = 3$ ; back-substitute remaining 4 eqs (e.g. Gaussian elim) for  $a_1..a_4$ . Non-singular  $\Rightarrow$  unique degree-4 poly.

**Free Parameters: 2 points, degree  $\leq 3$  (Quiz 4 Q3)**

Points  $(2, 5), (5, -1)$ :  $m + 1 = 2$ , so  $m = 1$ ; want degree  $n = 3$  (4 coeffs  $a_0, a_1, a_2, a_3$ ). Equations: 2. Free params =  $n - m = 3 - 1 = 2 = a_2, a_3$  free. **Construct the family:** constraints  $a_0 + 2a_1 + 4a_2 + 8a_3 = 5$  and  $a_0 + 5a_1 + 25a_2 + 125a_3 = -1$ . Subtract:  $3a_1 + 21a_2 + 117a_3 = -6 \Rightarrow a_1 = -2 - 7a_2 - 39a_3$ ; then  $a_0 = 9 + 10a_2 + 70a_3$ . **Family:**  $p(x) = (9 + 10a_2 + 70a_3) + (-2 - 7a_2 - 39a_3)x + a_2x^2 + a_3x^3$ , any  $a_2, a_3$ . Check:  $p(2) = 5$ ,  $p(5) = -1 \forall a_2, a_3$ . □

**Error calc:  $\sin x \approx x$  (Quiz 1)**

Exact  $f(x) = \sin x$ , approx  $\tilde{f}(x) = x$ . At  $x = 0.01$ :  $E_{abs} = |\sin(0.01) - 0.01| \approx |0.00999983 - 0.01| = 1.67 \times 10^{-7}$ ;  $E_{rel} = \frac{1.67 \times 10^{-7}}{\sin(0.01)} \approx 1.67 \times 10^{-5}$ . **MAE** over  $x = -0.01, 0.01, 0.03$ : errors  $1.67 \times 10^{-7}, 1.67 \times 10^{-7}, 4.50 \times 10^{-6}$ ; **MAE** =  $\frac{1.67 + 1.67 + 45.0}{3} \times 10^{-7} \approx 1.61 \times 10^{-6}$ . **RMSE** =  $\sqrt{\frac{(1.67)^2 + (1.67)^2 + (45.0)^2}{3}} \times 10^{-7} \approx 2.60 \times 10^{-6}$  (larger than MAE — squares penalize the big error).

**Truncation error bound example**

Generic bound formula applied literally with  $n = 8$ :  $f(x) = \sin(3x)$ , interval  $[-3, 4]$ ,  $x_0 = 0$ :  $f^{(9)}(x) = \pm 3^9 \cos(3x)$  or  $\pm 3^9 \sin(3x) \Rightarrow M = 3^9 = 19683$ .  $d = \max(|-3|, |4|) = 4$ .

$$|R_8(x)| \leq \frac{19683 \cdot 4^9}{9!} = \frac{19683 \cdot 262144}{362880} \approx 14219$$

(Large bound is fine — the point is applying the formula correctly, not that it's tight.)

**Quiz 3 Walkthrough (truncation error)**

**Key:**  $\sin(3x)$  odd  $\Rightarrow$  only odd powers; degree =  $2m + 1$  (even deg equals preceding odd).

**Q1:**  $\sin(3x) \simeq 3x - \frac{9x^3}{2} + \frac{81x^5}{40} - \frac{243x^7}{560} + \dots = \sum_k \frac{(-1)^k 3^{2k+1} x^{2k+1}}{(2k+1)!}$  (deg 8 stops at  $x^7$ ).

**Q2:** integral  $R_{2m+1} = \frac{(-1)^{m+1} 3^{2m+2}}{(2m+1)!} \int_0^x (x-t)^{2m+1} \sin(3t) dt$ ; Lagrange =  $\frac{(-1)^{m+1} 3^{2m+2} \sin(3\xi)}{(2m+2)!} x^{2m+2}$ .

**Q3:**  $|R_{2m+1}| \leq \frac{12^{2m+2}}{(2m+2)!}$  on  $[-3, 4]$ . **Q4:**

$m = 19 \Rightarrow \frac{12^{40}}{40!} \approx 1.80 \times 10^{-5}$  (fail);

$m = 20 \Rightarrow \frac{12^{42}}{42!} \approx 1.51 \times 10^{-6} \Rightarrow m = 20$  ( $2m + 2 = 42$ ).

**Q5:** even  $n = 8$  invalid for odd fn  $\Rightarrow n = 9$ :

$\frac{3^{10} x^{10}}{10!} < 10^{-7} \Rightarrow x < 0.3012$ ; or  $n = 7$ :  $\frac{3^8 x^8}{8!} \Rightarrow x < 0.1673$ .

**Maclaurin:  $\cos(4x)$  to degree 8 (HW06)**

$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \frac{u^6}{6!} + \frac{u^8}{8!} - \dots$ , so with  $u = 4x$ :  $\cos(4x) = 1 - 8x^2 + \frac{256}{24}x^4 - \frac{4096}{720}x^6 + \frac{65536}{40320}x^8 - \dots = 1 - 8x^2 + \frac{32}{3}x^4 - \frac{256}{45}x^6 + \frac{512}{315}x^8 - \dots$

**Pattern:**  $\cos(4x) = \sum_{k=0}^{\infty} \frac{(-1)^k (4x)^{2k}}{(2k)!}$ ; even powers only (even function), every other coefficient vanishes.  $M = 4^n + 1$  (amplitude of derivative).

**Taylor:  $\cos x - \sin x$  at  $\pi$  (HW05 Q4)**

$f(x) = \cos x - \sin x$ ,  $x_0 = \pi$ . Derivatives:  $f' = -\sin x - \cos x$ ,  $f'' = -\cos x + \sin x$ ,  $f''' = \sin x + \cos x$ . At  $\pi$ :  $f(\pi) = -1$ ,  $f'(\pi) = 1$ ,  $f''(\pi) = 1$ ,  $f'''(\pi) = -1$ .

$$T_1 = -1 + (x - \pi)$$

$$T_2 = -1 + (x - \pi) + \frac{1}{2}(x - \pi)^2$$

$$T_3 = T_2 - \frac{1}{6}(x - \pi)^3$$

**Key:** each new degree adds one term; recompute all derivatives at the *center*  $\pi$ , divide by  $k!$  (coefficients go  $-1, 1, 1, -1$ ).

**Taylor:  $\sin^2(2x)$  degree 4 at 0 (HW05 Q5)**

Use identity  $\sin^2(2x) = \frac{1}{2}(1 - \cos 4x)$ . From the  $\cos(4x)$  series:  $\sin^2(2x) = \frac{1}{2}\left(1 - (1 - 8x^2 + \frac{32}{3}x^4 - \dots)\right)$

$$\sin^2(2x) = 4x^2 - \frac{16}{3}x^4 + \dots \quad (\text{degree } 4)$$

**Or by derivatives:**  $f''(0) = 8 \Rightarrow \frac{8}{2!} = 4$ ;  $f^{(4)}(0) = -128 \Rightarrow \frac{-128}{4!} = -\frac{16}{3}$ . **Trap:**  $f(0) = f'(0) = f'''(0) = 0$  — don't invent odd/constant terms.

**Find  $n$  for tolerance:**  $\cos(4x)$ ,  $|R_n| < 0.001$  on  $[-3, 2]$  (HW06 Q4)

$M = 4^{n+1}$ ,  $d = \max|x| = 3$ , bound =  $\frac{4^{n+1} 3^{n+1}}{(n+1)!} = \frac{12^{n+1}}{(n+1)!}$ . Increase  $n$  until bound  $< 0.001$ :

$\frac{12^{36}}{36!} \approx 0.00191 > 0.001$ ,  $\frac{12^{37}}{37!} \approx 0.00062 < 0.001$   
 $\Rightarrow$  smallest  $n$  with  $|R_n| < 0.001$  is  $n = 36$ .

**Diff./Integration Quick Review**

Product rule:  $(fg)' = f'g + fg'$ . Quotient rule:

$$(f/g)' = \frac{f'g - fg'}{g^2}. \quad \text{Chain rule: } [f(g(x))]' = f'(g(x))g'(x).$$

Integration by parts:  $\int u dv =$

$uv - \int v du$ . **Example:**  $\int x \sin x dx$ : let  $u = x$ ,  $dv = \sin x dx \Rightarrow du = dx$ ,  $v = -\cos x$ .

$$\begin{aligned} \int x \sin x dx &= -x \cos x + \int \cos x dx \\ &= -x \cos x + \sin x + C \end{aligned}$$

**Centered Form: full worked example**

$p(x) = x^3 - 2x^2 + 3x - 5$ , center  $x_0 = 2$ . Repeatedly synthetic-divide by  $(x - 2)$ ; each remainder is the next  $b_k$  (coefficient of  $(x - x_0)^k$ ).

$$1, -2, 3, -5 \rightarrow 1, 0, 3 \mid b_0 = 1 \quad (\text{Div.1})$$

$$1, 0, 3 \rightarrow 1, 2 \mid b_1 = 7 \quad (\text{Div.2})$$

$$1, 2 \rightarrow 1 \mid b_2 = 4, b_3 = 1 \quad (\text{Div.3})$$

So  $p(x) = 1 + 7(x - 2) + 4(x - 2)^2 + (x - 2)^3$ . **Check:** expand back — matches original. □ Each division: bring down leading coeff, next = coeff +  $2 \times$  prev; last value = remainder.

**Standard  $\leftrightarrow$  Nested + FLOPs (HW03/Pract.3)**

$p_5(x) = x^5 + 3x^4 + x + 1$  (missing  $x^3, x^2 \Rightarrow$  insert 0's). Nested:

$$p_5(x) = 1 + x(1 + x(0 + x(0 + x(3 + x))))$$

**Verify:** matches at  $x = 0, 1, 2$ . ✓

**FLOPs (deg  $n$ ):** nested =  $n$  mult +  $n$  add =  $2n$  total ops; standard (recompute each power) = mults  $\frac{n(n+1)}{2}$  ( $\frac{n(n-1)}{2}$  for powers +  $n$  for coeffs) +  $n$  adds =  $\frac{n(n+3)}{2}$  total ops. For  $n = 5$ : nested = 10, standard = 20  $\Rightarrow$  **nested halves the work**.

**Horner: divide by  $ax - b$  (worked)**

Divide  $4x^3 - 7x^2 + 5x + 8$  by  $2x + 1$  (i.e.  $a = 2, b = -1$ , root  $c = -\frac{1}{2}$ ).

|                |   |    |     |       |
|----------------|---|----|-----|-------|
|                | 4 | -7 | 5   | 8     |
| $-\frac{1}{2}$ |   | -2 | 4.5 | -4.75 |
|                | 4 | -9 | 9.5 | 3.25  |

$q^*(x) = 4x^2 - 9x + 9.5$ , remainder  $r = 3.25$  (remainder unchanged by scaling). True quotient  $q(x) = q^*(x)/2 = 2x^2 - 4.5x + 4.75$ . **Check:**  $(2x + 1)(2x^2 - 4.5x + 4.75) + 3.25 = 4x^3 - 7x^2 + 5x + 8$ . ✓

**Bézier Curves: construction**

Bernstein form,  $n$  control pts  $P_0, \dots, P_n$ ,  $t \in [0, 1]$ :

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i P_i$$

Quadratic ( $n = 2$ ):  $B(t) = (1-t)^2 P_0 + 2t(1-t) P_1 + t^2 P_2$ . Cubic ( $n = 3$ ): use the full Bernstein sum above. **Key facts:** curve passes through  $P_0$  (at  $t = 0$ ) and  $P_n$  (at  $t = 1$ ); interior points *pull* the curve but aren't touched. **Worked example:** quadratic,  $P_0 = (0, 0)$ ,  $P_1 = (1, 2)$ ,  $P_2 = (3, 0)$ , find  $B(0.5)$ :

$$\begin{aligned} B(0.5) &= 0.25(0, 0) + 2(0.5)(0.5)(1, 2) + 0.25(3, 0) \\ &= (0, 0) + (0.5, 1) + (0.75, 0) = (1.25, 1) \end{aligned}$$

**Common Maclaurin Series (convergence)**

- $e^x = \sum \frac{x^k}{k!}$  (all  $\mathbb{R}$ )
- $\sin x = \sum \frac{(-1)^k x^{2k+1}}{(2k+1)!}$  (all  $\mathbb{R}$ )
- $\cos x = \sum \frac{(-1)^k x^{2k}}{(2k)!}$  (all  $\mathbb{R}$ )
- $\frac{1}{1-x} = \sum x^k$  ( $|x| < 1$ )
- $\ln(1+x) = \sum \frac{(-1)^{k+1} x^k}{k}$  ( $|x| < 1$ )
- $\tan^{-1} x = \sum \frac{(-1)^k x^{2k+1}}{2k+1}$  ( $|x| \leq 1$ )

**Trap:** writing the infinite sum doesn't prove  $f$  equals it — must show  $R_n \rightarrow 0$  (see §4).